Of course, ekpyrotic models are not the only way to put some bounce in our cosmic step.

Roger Penrose, an early pioneer of modern cosmology who fundamentally changed the way we look at gravity in the universe, has his own proposal for a cycling cosmos, in which our Big Bang is born from a previous cycle's Heat Death. It involves piecing together the far future spacetime of one universe and the singularity at the beginning of another. Penrose has been, for decades, one of the most prominent voices in cosmology pointing out the seriousness of the entropy problem in standard early universe scenarios. And he does *not* think inflation does the trick. He told me recently, "When I first heard about it, I thought, well, that theory won't last a week."

Penrose's alternative model, called Conformal Cyclic Cosmology, conjectures that entropy works differently in the vicinity of singularities. If the conjecture is true, it implies that the entropy would be very low at the boundary between cycles from which our universe begins, and it doesn't require inflation. Penrose's model also contains the intriguing possibility that some imprint of the events that occurred in past cycles might appear in astronomical observations, showing up as features in the cosmic microwave background. In fact, Penrose and his collaborators have claimed that evidence for such features can already be seen in the data, though this has been met with skepticism. Whether or not these possible CMB hints will someday be seen as a compelling sign of a pre–Big Bang universe is yet to be determined.

Meanwhile, ekpyrotic-model co-developer Neil Turok has shifted focus to dive into a totally new model of the universe in which the Big Bang is merely a transition point. This proposal, developed by Latham Boyle, Turok, and their former student Kieran Finn, is motivated by taking symmetry arguments in particle physics to a cosmic level; it suggests that our universe and a time-reversed version of the cosmos meet at the Big Bang like two cones touching tip to tip. In a recent paper, they describe the picture as "a universe-antiuniverse pair, emerging from nothing." It's possible that cone-tip singularity might contain its own solution to the entropy problem, though the model and its details are (at time of this writing) still under development. Nonetheless, it makes some specific predictions for the nature of dark matter, and thus might be testable with upcoming experiments.